

Manan Joshi

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Portfolio Website

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[GeeksforGeeks](#) [CodeChef](#)

EDUCATION

Dr. B.R. Ambedkar National Institute of Technology, Jalandhar

2023 - Current

B.Tech - Information Technology - CGPA - 8.28

Jalandhar, Punjab

EXPERIENCE

C++ Programming Intern | CodeAlpha (Remote)

May 2025 - Jun 2025

- Accelerated the delivery of 4 C++ projects by abstracting complex logic into reusable OOP classes, reducing code redundancy by 40%.
- Programmed a file-based authentication mechanism for 100+ users, utilizing Regex for strict input validation to prevent data corruption.
- Constructed a secure, modular Banking System with transactions and record tracking for 100+ users.
- Refined modular codebase in a remote setup, cutting development errors by 25% via early-stage bug tracking.

PROJECTS

Aegis: Autonomous AI Productivity Engine (MERN) [↗](#)

Jun 2026

- Traditional productivity tools exacerbate executive dysfunction through passive notifications; designed an active intervention engine to automate distraction-free task execution.
- Orchestrated a multi-agent routing layer via Google Gemini (gemini-2.5-flash) with strict JSON schema enforcement, achieving 100% parsing accuracy across 3 specialized AI personas during autonomous task decomposition.
- Developed a custom mathematical Urgency Engine in Node.js to evaluate real-time U-Scores (Time Remaining vs. Estimated Time) in under 50ms, dynamically driving UI state changes and workflow escalations.
- Built a state-driven React and Zustand frontend incorporating a visually isolated Focus Tunnel alongside native Web Speech and SpeechSynthesis APIs for 100% hands-free bi-directional voice accessibility.

Tech Stack: React.js, Node.js, Express.js, MongoDB, Google Gemini SDK, Google Cloud Run, JWT Authentication, Zustand, Web Speech API, Tailwind CSS

FinTrack: Personal Finance Dashboard (MERN) [↗](#)

Feb 2026

- Manual financial tracking lacks predictive forecasting and relies on rigid, error-prone data entry, creating friction in personal expense management.
- Architected a dedicated Express.js backend to decouple API routing, integrating Mindee AI to asynchronously parse unstructured receipt payloads in under 2 seconds, reducing manual data entry time by ~90%.
- Engineered a computationally heavy state engine utilizing a custom 3-month moving average algorithm to analyze historical ledger data, forecast baseline expenses, and dynamically execute 50-30-20 budget reallocations.
- Secured 15+ critical REST API endpoints utilizing stateless JWT payloads, ensuring zero-exposure environments for sensitive user ledger data while fully automating the financial tracking pipeline.

Tech Stack: React.js, Node.js, Express.js, MongoDB, Tailwind CSS, Mindee AI, JWT Authentication

TECHNICAL SKILLS

- Languages: C++, C, Python, JavaScript, SQL, HTML/CSS
- Libraries/Frameworks: React.js, Node.js, Express.js, Next.js, Tailwind CSS, Mongoose, Zustand
- Tools: Git/GitHub, VS Code, MongoDB Compass, REST API, Postman, Figma, JWT Authentication, Google Gemini SDK (GenAI)
- Coursework: Data Structures & Algorithms, OOP, DBMS, OS, Computer Networks

ACHIEVEMENTS

- National Engineering Competitor: Advanced as a National Semi-Finalist in Flipkart GRID 7.0 (Software Development Track) and ranked Top 1500 nationally in Google The Big Code 2026 out of thousands of participants.
- Algorithmic Proficiency: Attained LeetCode Guardian (2200+) | CodeChef 4-Star (1800+) | Codeforces Specialist (1400+) | Achieved Institute Rank 1 on GeeksforGeeks by mastering core data structures.
- Applied AI & Advanced Logic: Certified in Google & Kaggle AI Agents for architecting autonomous workflows, and mastered 17 advanced graph theory paradigms under a Codeforces Master.